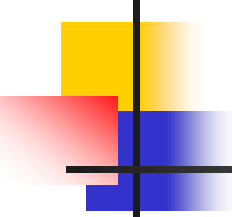


Introduction to Time- Frequency Analysis Techniques



Outline of Talk

- Overview
- Historical Development
- Time vs Frequency Domain Analysis
- Fourier Analysis
- Fourier vs Wavelet Transforms
- Wavelet Analysis
- Typical Applications
- References



OVERVIEW

- Wavelet
 - A small wave
- Wavelet Transforms
 - Convert a signal into a series of wavelets
 - Provide a way for analyzing waveforms, bounded in both frequency and duration
 - Allow signals to be stored more efficiently than by Fourier transform
 - Be able to better approximate real-world signals
 - Well-suited for approximating data with sharp discontinuities
- “The Forest & the Trees”
 - Notice gross features with a large “window”
 - Notice small features with a small



Historical Development

- Pre-1930
 - Joseph Fourier (1807) with his theories of frequency analysis
- The 1930^s
 - Using scale-varying basis functions; computing the energy of a function
- 1960-1980
 - Guido Weiss and Ronald R. Coifman; Grossman and Morlet
- Post-1980
 - Stephane Mallat; Y. Meyer; Ingrid Daubechies; wavelet applications today



Mathematical Transformation

- Why
 - To obtain a further information from the signal that is not readily available in the raw signal.
- Raw Signal
 - Normally the time-domain signal
- Processed Signal
 - A signal that has been "transformed" by any of the available mathematical transformations
- Fourier Transformation
 - The most popular transformation



FREQUENCY ANALYSIS

- Frequency Spectrum
 - Be basically the frequency components (spectral components) of that signal
 - Show what frequencies exists in the signal
- Fourier Transform (FT)
 - One way to find the frequency content
 - Tells how much of each frequency exists in a signal

$$\begin{aligned}X(k+1) &= \sum_{n=0}^{N-1} x(n+1) \cdot W_N^{kn} \\x(n+1) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k+1) \cdot W_N^{-kn} \\w_N &= e^{-j\left(\frac{2\pi}{N}\right)}\end{aligned}$$

$$\begin{aligned}X(f) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-2j\pi ft} dt \\x(t) &= \int_{-\infty}^{\infty} X(f) \cdot e^{2j\pi ft} df\end{aligned}$$



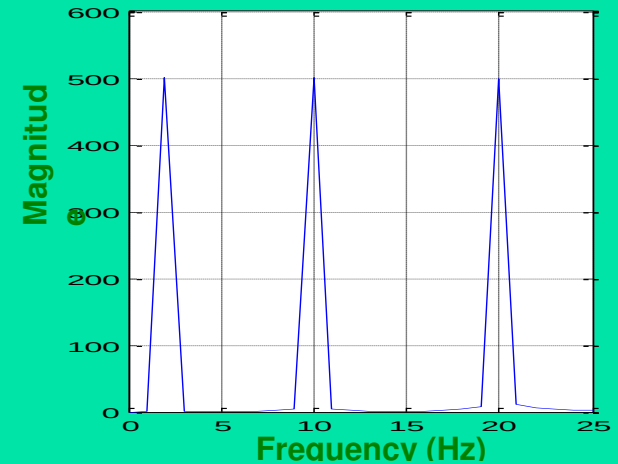
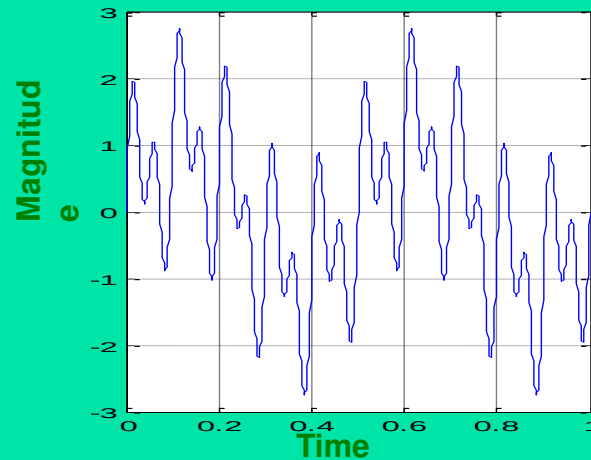
STATIONARITY OF SIGNAL

- Stationary Signal
 - Signals with frequency content unchanged in time
 - All frequency components exist at all times
- Non-stationary Signal
 - Frequency changes in time
 - One example: the “Chirp Signal”

STATIONARITY OF SIGNAL

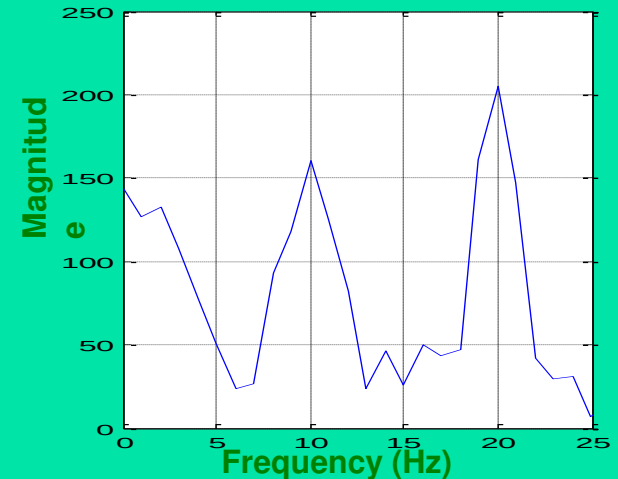
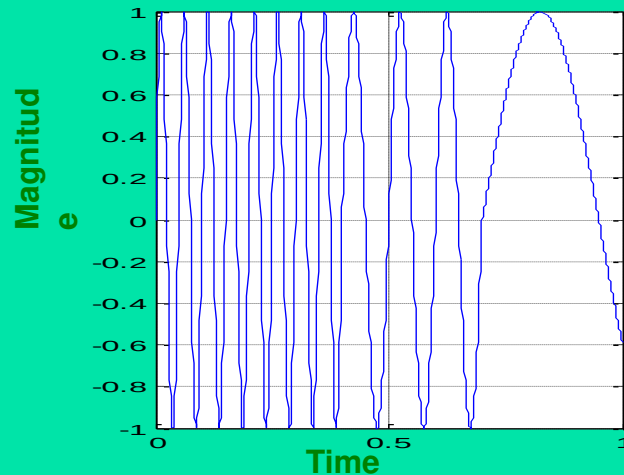
2 Hz + 10 Hz + 20Hz

Stationary



0.0-0.4: 2 Hz +
0.4-0.7: 10 Hz +
0.7-1.0: 20Hz

Non-Stationary

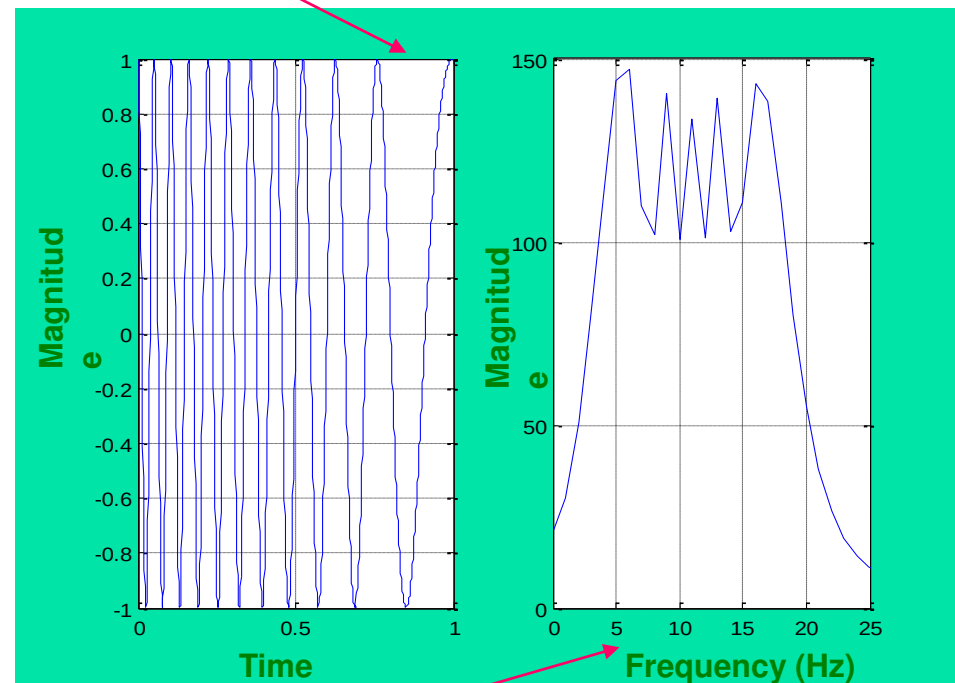
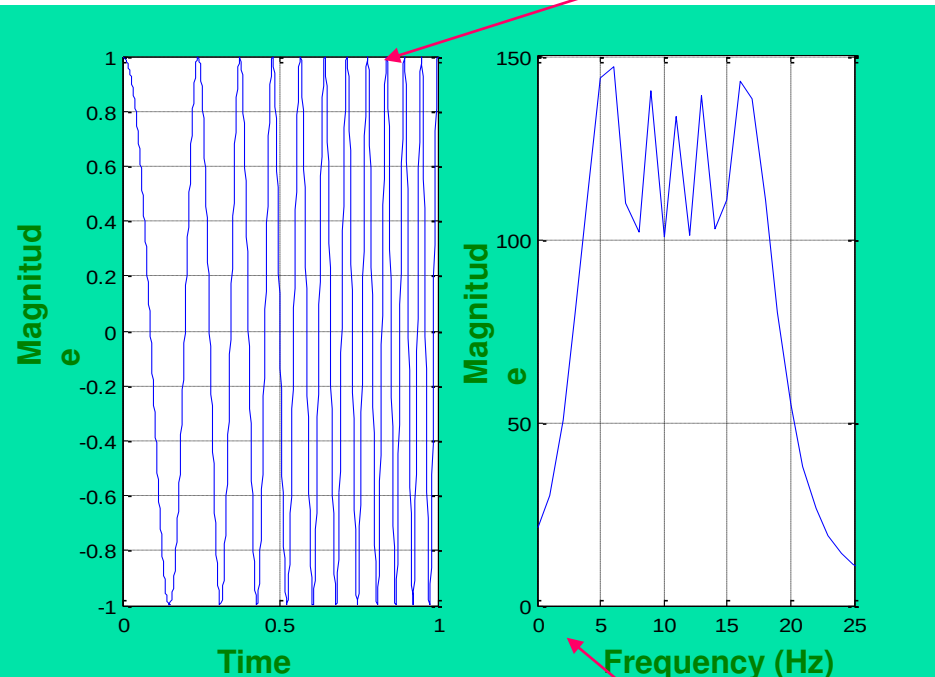


CHIRP SIGNALS

Frequency: 2 Hz to 20 Hz

Different in Time Domain

Frequency: 20 Hz to 2 Hz



Same in Frequency Domain

At what time the frequency components occur? FT can not tell!



NOTHING MORE, NOTHING LESS

- FT Only Gives what Frequency Components Exist in the Signal
- The Time and Frequency Information can not be Seen at the Same Time
- Time-frequency Representation of the Signal is Needed

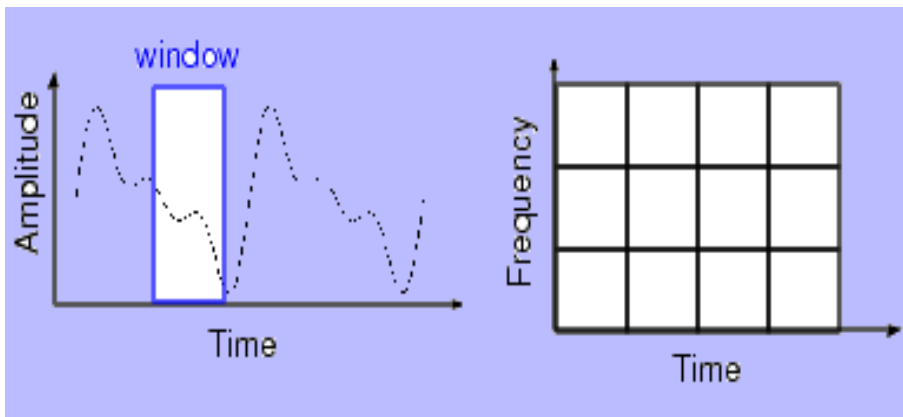
Most of Transportation Signals are Non-stationary.

(We need to know **whether** and also **when** an incident was happened.)

ONE EARLIER SOLUTION: SHORT-TIME FOURIER TRANSFORM (STFT)

SHORT TIME FOURIER TRANSFORM (STFT)

- Dennis Gabor (1946) Used STFT
 - To analyze only a small section of the signal at a time -- a technique called *Windowing the Signal*.
- The Segment of Signal is Assumed *Stationary*
- A 3D transform



$$\text{STFT}_X^{(\omega)}(t', f) = \int [x(t) \bullet \omega^*(t - t')] \bullet e^{-j2\pi ft} dt$$

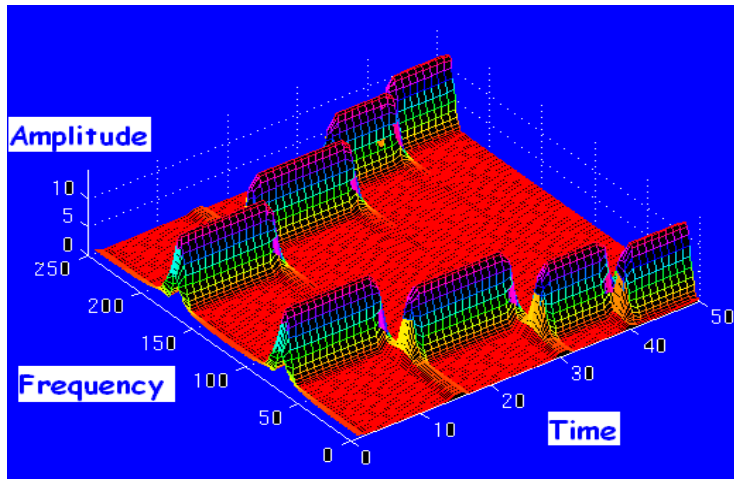
$\omega(t)$: the window function

**A function of time
and frequency**

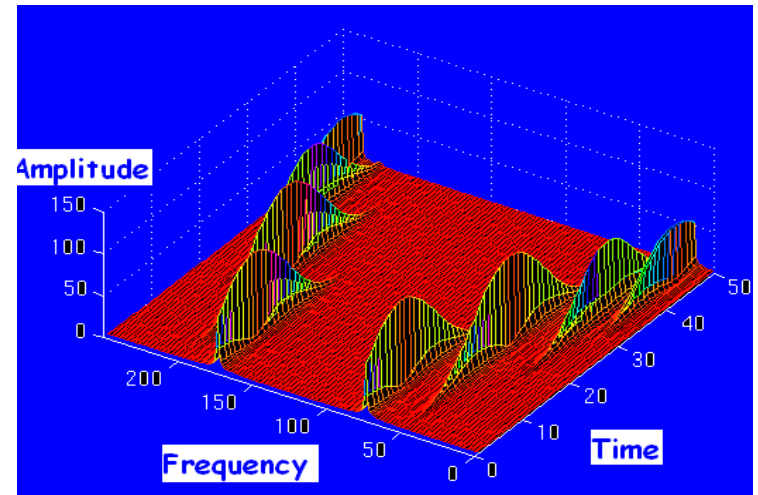
DRAWBACKS OF STFT

- Unchanged Window
- Dilemma of Resolution
 - Narrow window -> poor frequency resolution
 - Wide window -> poor time resolution
- Heisenberg Uncertainty Principle
 - Cannot know what frequency exists at what time intervals

Via Narrow Window



Via Wide Window





MULTIRESOLUTION ANALYSIS (MRA)

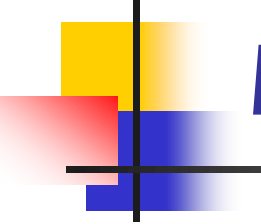
- Wavelet Transform
 - An alternative approach to the short time Fourier transform to overcome the resolution problem
 - Similar to STFT: signal is multiplied with a function
- Multiresolution Analysis
 - Analyze the signal at different frequencies with different resolutions
 - Good time resolution and poor frequency resolution at high frequencies
 - Good frequency resolution and poor time resolution at low frequencies
 - More suitable for short duration of higher frequency; and longer duration of lower frequency components



PRINCIPLES OF WAVELET TRANSFORM

- Split Up the Signal into a Bunch of Signals
- Representing the Same Signal, but all Corresponding to Different Frequency Bands
- Only Providing What Frequency Bands Exists at What Time Intervals

DEFINITION OF CONTINUOUS WAVELET TRANSFORM


$$\text{CWT}_x^\psi(\tau, s) = \Psi_x^\psi(\tau, s) = \frac{1}{\sqrt{|s|}} \int x(t) \bullet \psi^*\left(\frac{t - \tau}{s}\right) dt$$

Translation

(The location of the window)

Scale

Mother Wavelet

- Wavelet
 - Small wave
 - Means the window function is of finite length
- Mother Wavelet
 - A prototype for generating the other window functions
 - All the used windows are its dilated or compressed and shifted versions

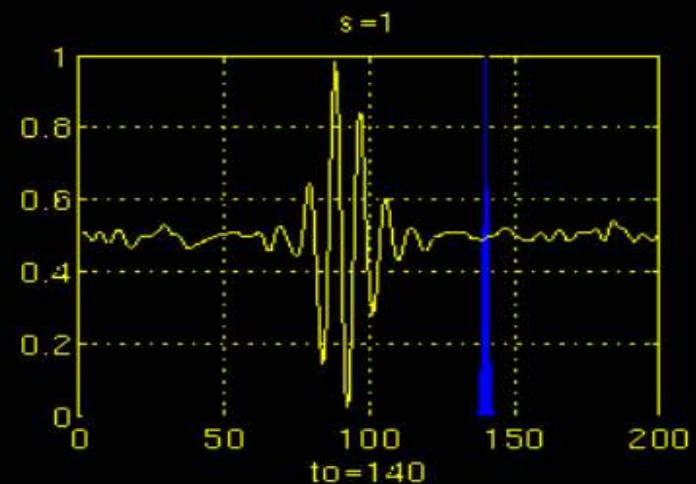
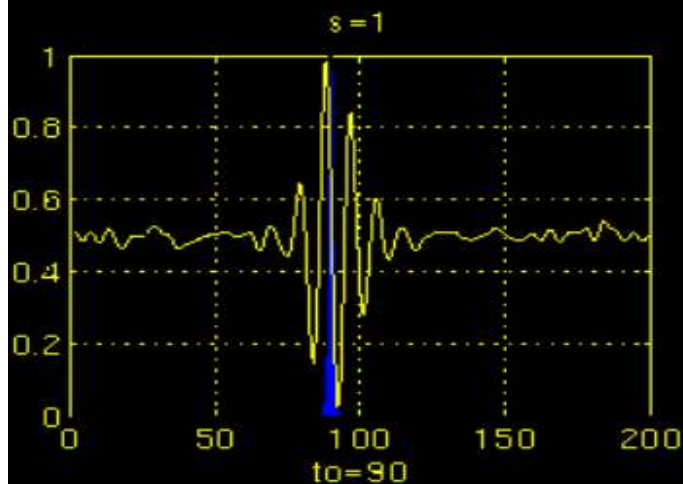
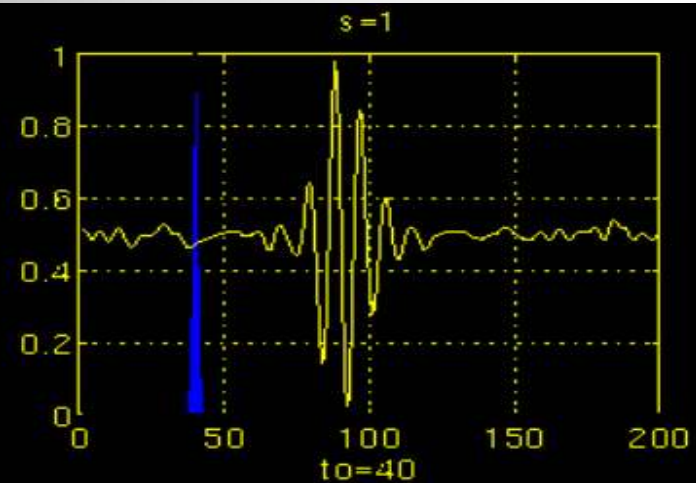
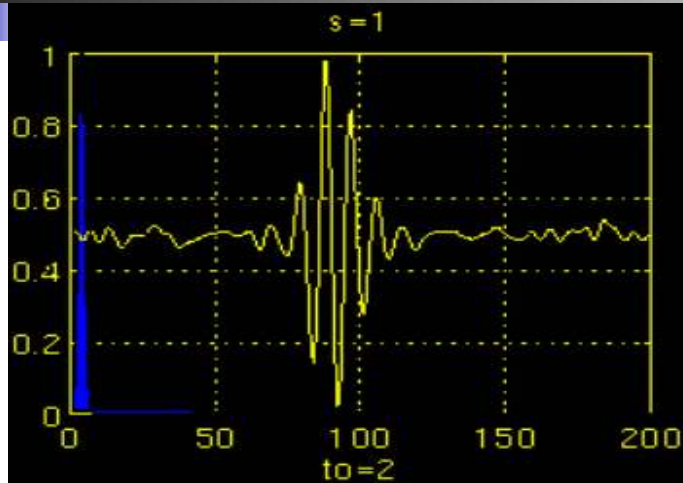


SCALE

- Scale
 - $S > 1$: dilate the signal
 - $S < 1$: compress the signal
- Low Frequency -> High Scale -> Non-detailed
Global View of Signal -> Span Entire Signal
- High Frequency -> Low Scale -> Detailed
View Last in Short Time
- Only Limited Interval of Scales is Necessary

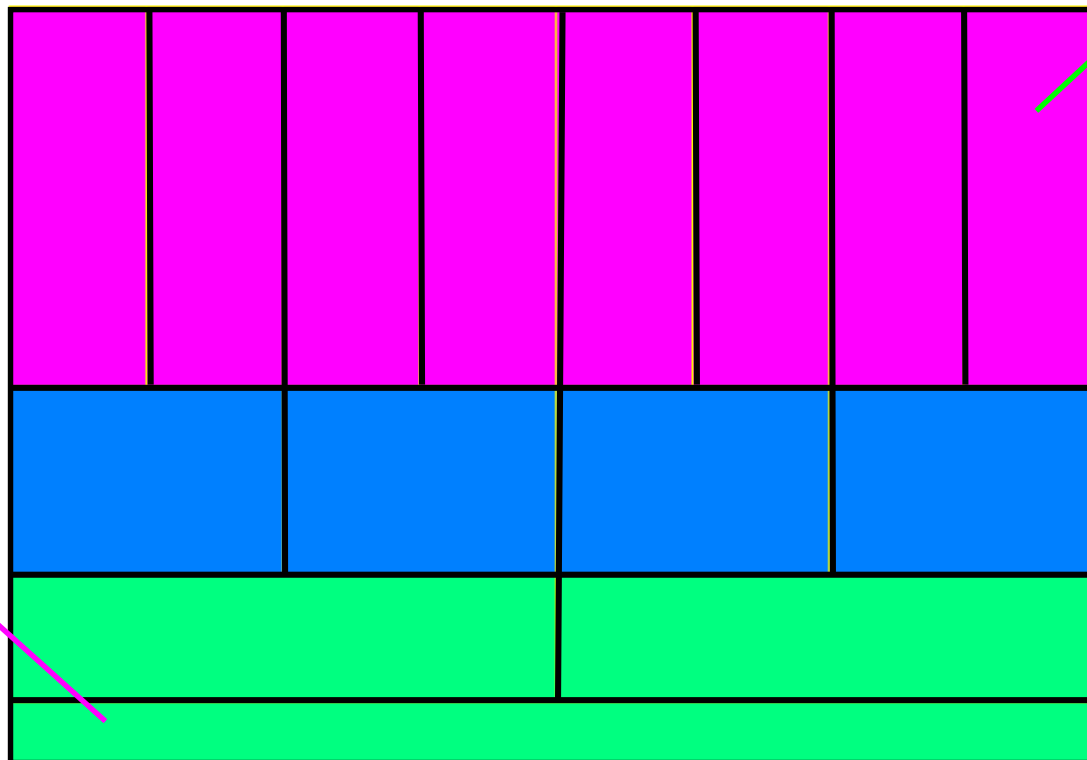
$$\text{CWT}_x^\Psi(\tau, s) = \Psi_x^\Psi(\tau, s) = \frac{1}{\sqrt{|s|}} \int x(t) \bullet \psi^*\left(\frac{t - \tau}{s}\right) dt$$

COMPUTATION OF CWT



RESOLUTION OF TIME & FREQUENCY

Frequency



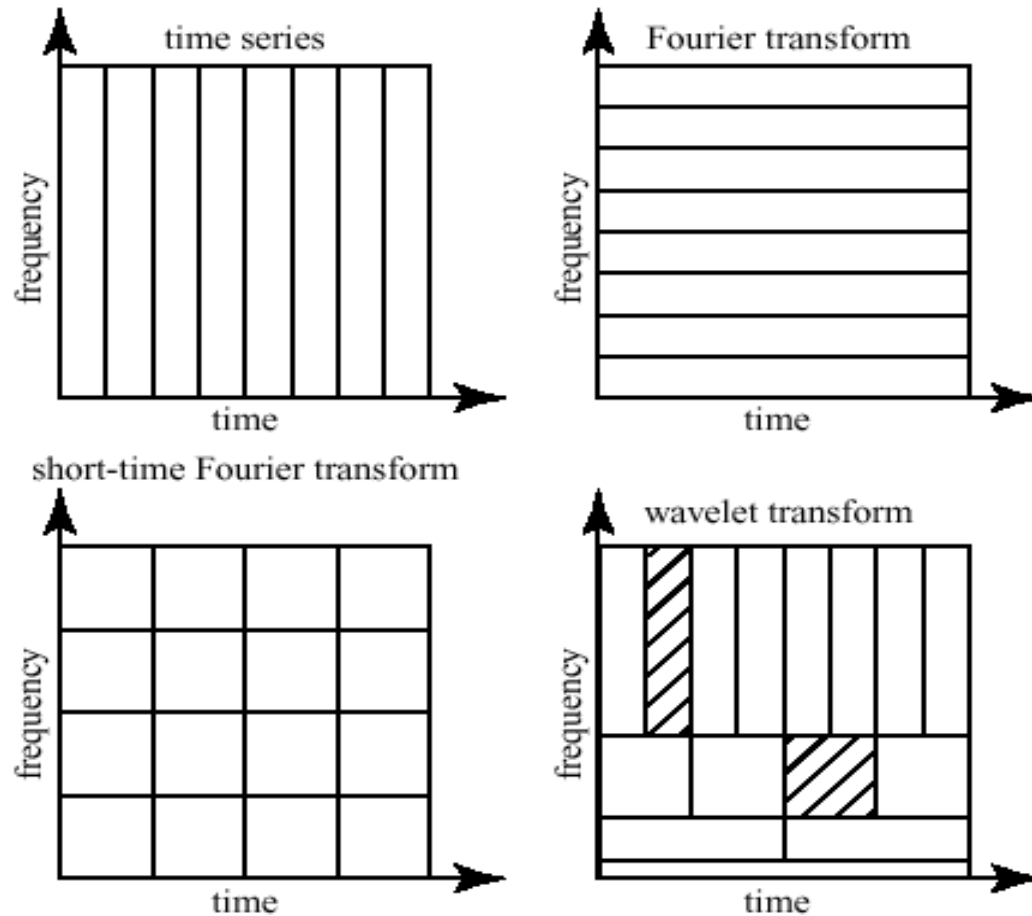
**Better time
resolution;
Poor
frequency
resolution**

**Better
frequency
resolution;
Poor time
resolution**

Time

- Each box represents an equal portion
- Resolution in STFT is selected once for entire analysis

COMPARSION OF TRANSFORMATIONS





DISCRETIZATION OF CWT

- It is Necessary to Sample the Time-Frequency (scale) Plane.
- At High Scale s (Lower Frequency f), the Sampling Rate N can be Decreased.
- The Scale Parameter s is Normally Discretized on a Logarithmic Grid.
- The most Common Value is 2.
- The Discretized CWT is not a True Discrete Transform

- Discrete Wavelet Transform (DWT)
 - Provides sufficient information both for analysis and synthesis
 - Reduce the computation time sufficiently
 - Easier to implement
 - Analyze the signal at different frequency bands with different resolutions
 - Decompose the signal into a coarse approximation and detail information

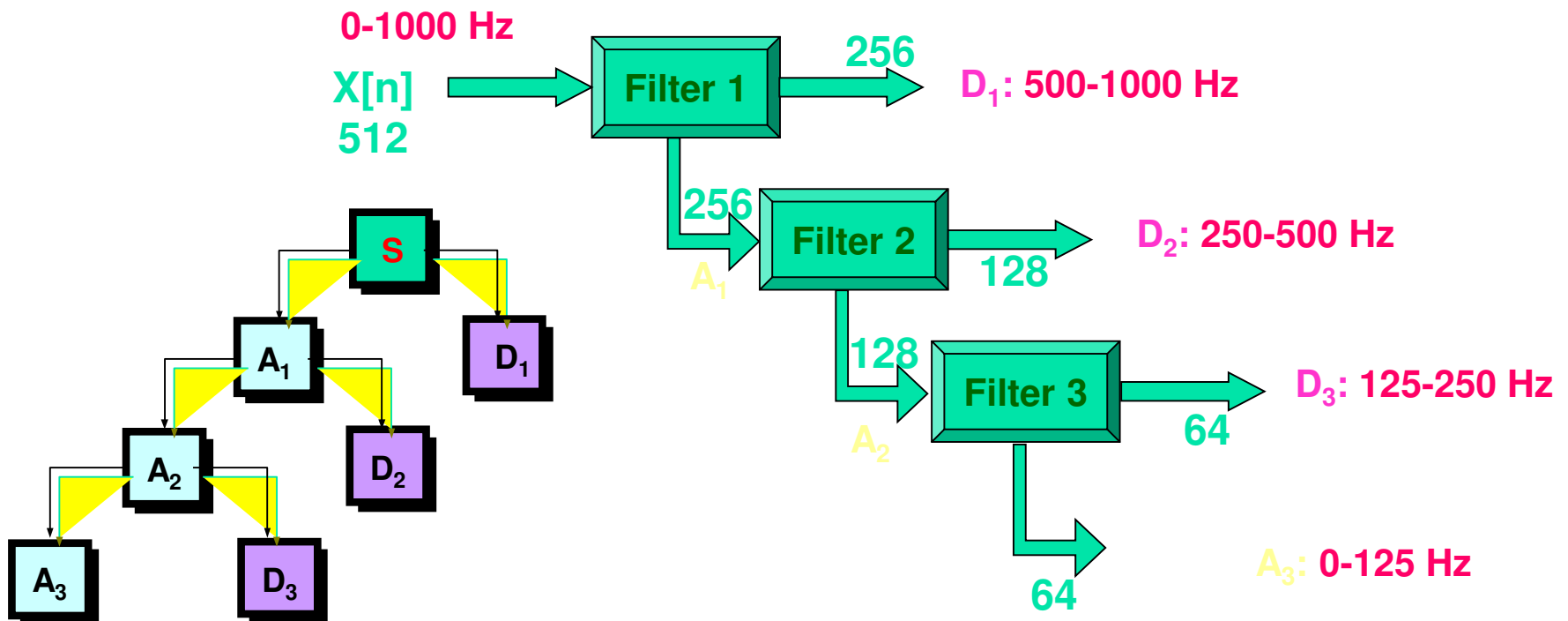


Multi Resolution Analysis

- Analyzing a signal both in time domain and frequency domain is needed many a times
 - But resolutions in both domains is limited by Heisenberg uncertainty principle
- Analysis (MRA) overcomes this , how?
 - Gives good time resolution and poor frequency resolution at high frequencies and good frequency resolution and poor time resolution at low frequencies
 - This helps as most natural signals have low frequency content spread over long duration and high frequency content for short durations

SUBBABD CODING ALGORITHM

- **Halves the Time Resolution**
 - Only half number of samples resulted
- **Doubles the Frequency Resolution**
 - The spanned frequency band halved





RECONSTRUCTION

- What
 - How those components can be assembled back into the original signal without loss of information?
 - A Process After *decomposition* or *analysis*.
 - Also called *synthesis*
- How
 - Reconstruct the signal from the wavelet coefficients
 - Where wavelet analysis involves filtering and down sampling, the wavelet reconstruction process consists of up sampling and filtering



WAVELET APPLICATIONS

- Typical Application Fields
 - Astronomy, acoustics, nuclear engineering, sub-band coding, signal and image processing, neurophysiology, music, magnetic resonance imaging, speech discrimination, optics, fractals, turbulence, earthquake-prediction, radar, human vision, and pure mathematics applications
- Sample Applications
 - Identifying pure frequencies
 - De-noising signals
 - Detecting discontinuities and breakdown points
 - Detecting self-similarity
 - Compressing images